Homework 7

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Physics 15
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Question 1: ER 5.7. Use the particle in a box wave function verified in Example 5-9, with the value of A determined in Example 5-10, to calculate the probability that the particle associated with the wave function would be found in a measurement within a distance of $a/3$ from the right-hand end of the box of length a. The particle is in its lowest energy state. (b) Compare with the probability that would be predicted classically from a very simple calculation related to the one in Example 5-6.

Solution. Recall that from Example 5-10 that

$$\Psi(x,t) = \sqrt{\frac{2}{a}} \cos\left(\frac{\pi x}{a}\right) e^{-i\omega t} \implies \psi = \sqrt{\frac{2}{a}} \cos\left(\frac{\pi x}{a}\right).$$

To find the probability from the right third of the particle well, let us first find the the boundary conditions of the integral. Since the well is centered at $x = 0$ with length of $L = a$, the right third segment of the well is when $a/6 \leq x \leq a/2$. Therefore, probability p is

$$\begin{aligned}
p &= \int_{a/6}^{a/2} |\psi|^2 dx = \frac{2}{a} \int_{a/6}^{a/2} \cos^2\left(\frac{\pi x}{a}\right) dx \\
&= \frac{1}{a} \int_{a/6}^{a/2} 1 + \cos\left(\frac{2\pi x}{a}\right) dx = \frac{1}{a}\left[x + \frac{a}{2\pi}\sin\left(\frac{2\pi x}{a}\right) - x\right]_{x=a/6}^{x=a/2} \\
&= \frac{1}{a}\left[\left(\frac{a}{2} - \frac{a}{6}\right) + \frac{a}{2\pi}\left(\sin\pi - \sin\frac{\pi}{3}\right)\right] = \frac{1}{3} - \frac{\sqrt{3}}{4\pi} \approx \boxed{0.195.}
\end{aligned}$$

This drastically differs from the classical predictions, where the probability p' is the ratio of the length of the region over the total region, which is

$$p' = \frac{a/3}{a} = \boxed{\frac{1}{3}.}$$

□

Question 2: ER 5.23. Consider a particle moving in the potential $V(x)$ plotted in Figure 5-22. For the following ranges of the total energy E, state whether there are any allowed values of E and if so, whether they are discretely separated or continuously distributed. (a) $E < V_0$, (b) $V_0 < E < V_1$, (c) $V_1 < E < V_2$, (d) $V_2 < E < V_3$, (e) $V_3 < E$.

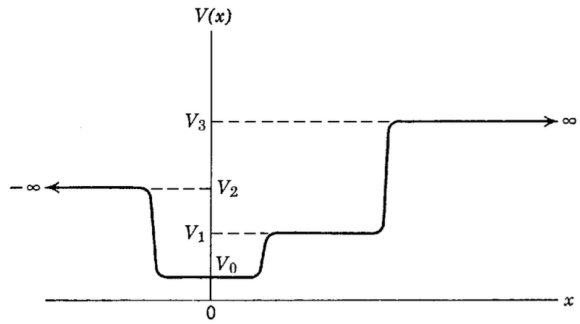


Figure 5-22 A potential function considered in Problem 23.

Solution. For part (a), *there are no allowed values of* E. Because $E < V_0$, the total energy is less than the minimum potential energy of the system, meaning the kinetic energy would have to be negative, which is physically impossible.

For part (b), the allowed values of E are *discretely separated.* In this range, the particle is confined within the deepest potential well. Since it is trapped between two boundaries higher than its energy, it forms bound states with quantized energy levels.

Similarly, for part (c), the allowed values of E are *discretely separated.* Although the particle can now access a wider region of space to the right, it is still confined by the potential barriers V_2 on the left and V_3 on the right.

However, for part (d), the allowed values of E are *continuously distributed.* Because $E > V_2$, the particle is no longer confined on the left side and can escape to $x = -\infty$.

Similarly, for part (e), the allowed values of E are *continuously distributed.* At this energy level, E is greater than all the potential barriers shown in the figure. □

Question 3. A Gaussian wave packet is given by

$$\psi(x) = \left(\frac{1}{a^2\pi}\right)^{1/4} e^{-x^2/2a^2}$$

(a) Sketch the wave packet (amplitude as a function of x).

(b) Calculate the expectation values $\langle x\rangle$, $\langle x^2\rangle$, $\langle p\rangle$, and $\langle p^2\rangle$.

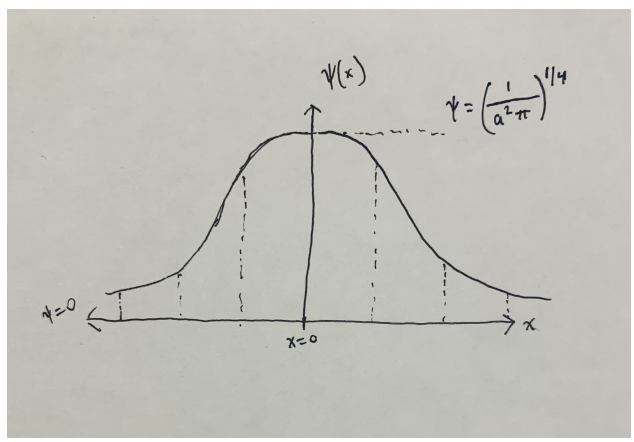
(c) Find the uncertainty relation for a Gaussian wave packet, i.e. find the product $\Delta x \Delta p$.

(d) The energy associated with this wave function is

$$E = \frac{\hbar^2}{2ma^2}$$

Find the potential energy as a function of x (Hint: use the time-independent Schrödinger Equation!). What is the classical potential that has this x-dependence?

Solution. We have sketched the wave packet as such below.



For part (b), let us find the following expectation values: $\langle x\rangle, \langle x^2\rangle, \langle p\rangle,$ and $\langle p^2\rangle$.
For $\langle x\rangle$,

$$\langle x\rangle = \int_{-\infty}^{\infty} \psi^*[x]\psi dx = \int_{-\infty}^{\infty} x|\psi|^2 dx.$$

However, we know that $y = \left(\frac{1}{a^2\pi}\right)^{1/4} e^{-x^2/2a^2}$ is an even function while $y = x$ is an odd function. Hence, $y = x|\psi|^2$ is an odd function, and $\boxed{\langle x\rangle = 0.}$

We can use this similar logic to find $\langle x^2\rangle$. Since $y = x^2$ is an even function, $y = x^2|\psi|^2$ is also even, and

$$\langle x^2\rangle = \int_{-\infty}^{\infty} x^2|\psi|^2 dx = \int_{-\infty}^{\infty} x^2 \left[\left(\frac{1}{a^2\pi}\right)^{1/4} e^{-x^2/2a^2}\right]^2 dx = \sqrt{\frac{1}{a^2\pi}} \int_{-\infty}^{\infty} x^2 e^{-x^2/a^2} dx$$

$$= \sqrt{\frac{1}{a^2\pi}} \left(\frac{1}{2(1/a^2)}\right) \sqrt{\frac{\pi}{1/a^2}} = \boxed{\frac{a^2}{2}}.$$

Let us now evaluate $\langle p \rangle$. Since the wavefunction is a symmetric bell curve about $x = 0$, implying that the probability of the wave packet moving to the left is equal to the probability of the packet moving to the right. Therefore, these probabilities cancel out, implying $\boxed{\langle p \rangle = 0.}$ Now, let us find $\langle p^2 \rangle$:

$$\langle p^2 \rangle = \int_{-\infty}^{\infty} \psi^* \left[i\hbar \frac{d}{dx} \right]^2 \psi dx = -\hbar^2 \int_{-\infty}^{\infty} \psi \frac{d^2\psi}{dx^2} dx.$$

Notice, when setting $\alpha = e^{-x^2/2a^2}$, that

$$\begin{aligned} \frac{d^2}{dx^2}(e^{-x^2/2a^2}) &= \frac{d^2}{dx^2}\alpha = \frac{d}{dx}\left(\frac{-2x}{2a^2} e^{-x^2/2a^2} \right) = -\frac{1}{a^2}\frac{d}{dx}(x\alpha) \\ &= -\frac{1}{a^2}(\alpha + x\frac{d}{dx}\alpha) = -\frac{1}{a^2}(\alpha - \frac{x^2}{a^2}\alpha) = \frac{x^2}{a^4}\alpha - \frac{1}{a^2}\alpha. \end{aligned}$$

Therefore, $\frac{d^2\psi}{dx^2} = \frac{x^2\psi}{a^4} - \frac{\psi}{a^2}$, and substituting this into the original integral gives us

$$\begin{aligned} \langle p^2 \rangle &= -\hbar^2 \int_{-\infty}^{\infty} \psi \left(\frac{x^2}{a^4} - \frac{1}{a^2} \right) \psi dx = -\hbar^2 \left[\frac{1}{a^4} \int_{-\infty}^{\infty} x^2 |\psi|^2 dx - \frac{1}{a^2} \int_{-\infty}^{\infty} |\psi|^2 dx \right] \\ &= -\hbar^2 \left[\frac{1}{a^4} \left(\frac{a^2}{2} \right) - \frac{1}{a^2} \right] = -\hbar^2 \left(-\frac{1}{2a^2} \right) = \boxed{\frac{\hbar^2}{2a^2}.} \end{aligned}$$

For part (c), since $\langle x \rangle = \langle p \rangle = 0$, the uncertainties Δx and Δp are simply

$$\Delta x = \sqrt{\langle x^2 \rangle} = \frac{a}{\sqrt{2}}, \quad \Delta p = \sqrt{\langle p^2 \rangle} = \frac{\hbar}{a\sqrt{2}}.$$

Hence,

$$\Delta x \Delta p = \left(\frac{\cancel{a}}{\sqrt{2}} \right) \left(\frac{\hbar}{\cancel{a}\sqrt{2}} \right) = \boxed{\frac{\hbar}{2}.}$$

This satisfies Heisenberg's Uncertainty Principle as expected! For part (d), let us implement the time-independent Schrödinger Equation:

$$\frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi.$$

By substitution of $\frac{d^2\psi}{dx^2} = \frac{x^2\psi}{a^4} - \frac{\psi}{a^2}$ and $E = \hbar^2/2ma^2$, we reduce the equation so that

$$\frac{-\hbar^2}{2m}\left(\frac{x^2\cancel{\psi}}{a^4} - \frac{\cancel{\psi}}{\cancel{a^2}} \right) + V\cancel{\psi} = \cancel{\frac{\hbar^2}{2ma^2}\psi} \implies \boxed{V = \frac{\hbar^2 x^2}{2ma^4}.}$$

This potential bears sembelance to the potential of the spring $U = \frac{1}{2}kx^2$. By this analogy, the "spring constant" of the potential in this wave packet would be

$$\boxed{k_{\text{eff}} = \frac{\hbar^2}{ma^4}.}$$

$\square$